DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
University of Rajshahi
B.Sc. Engineering Part III Even Semester Exam -2023
CSE 3252: Computer Networks Lab.

Set-A

Roll No: **Time: 03 Hours**

[Solve the marked two]

1. a) Building Ethernet cable (straight) and testing.
   b) Find the IP Address of your PC.
   c) Set the IP address of PC1: 192.168.0.1 and PC2: 192.168.0.2 – ping from PC1 to PC2.
   d) Share a folder in PC1 and access it from PC2.
2. Write a program to send an email from your university account to asif@ru.ac.bd. The email should have the following info:
   - Subject: Exam Mail: Your_Roll_Number
   - Body: your machine's IP address.
     You can use the following method to get the local machine's IP address.
     java.net.InetAddress.getLocalHost().getHostAddress()
   - Your system time.
3. **Packet Tracer Problem Statement: Inter-VLAN Routing Using Router-on-a-Stick**

Objective: Design and simulate a small network in Cisco Packet Tracer where two manageable switches and a single router are used to implement inter-VLAN routing.

Network Setup:

You are provided with the following devices:

- **2 Manageable Switches** (Switch1 and Switch2)
- **1 Router** (Router0)
- **10 PCs** connected to each switch (total 20 PCs)

VLAN Configuration:

- On both **Switch1** and **Switch2**:
  - **Ports Fa0/1 to Fa0/5** should be assigned to **VLAN 101**
  - **Ports Fa0/6 to Fa0/10** should be assigned to **VLAN 102**

Requirements:

1. **Configure VLANs** on both switches.
2. Assign **PCs in ports 1–5** to VLAN 101 and **ports 6–10** to VLAN 102.
3. Configure **trunk links** between:
   - Each switch and the router
   - (Optional) Between Switch1 and Switch2 (if needed for full connectivity)
4. Configure **subinterfaces** on the router for inter-VLAN routing.
5. Assign proper IP addresses to all PCs based on their VLAN.
6. **Verify inter-VLAN communication**, i.e., a PC in VLAN 101 should be able to ping a PC in VLAN 102 across switches.
7. Successful **ping tests** between PCs in different VLANs (e.g., PC0 in VLAN 101 and PC9 in VLAN 102).

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Set-B

Roll No: **Time: 03 Hours**

[Solve the marked two]

1. a) Building Ethernet cable (straight) and testing.
   b) Find the IP Address of your PC.
   c) Set the IP address of PC1: 192.168.0.1 and PC2: 192.168.0.2 – ping from PC1 to PC2.
   d) Share a folder in PC1 and access it from PC2.
2. Write a client-server program to implement the following protocol:

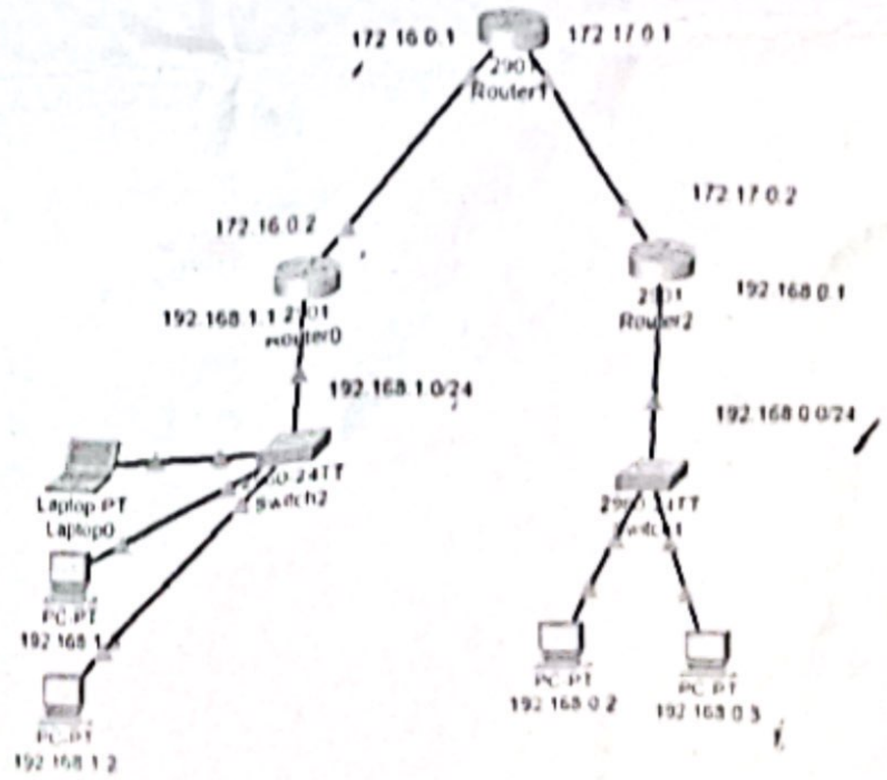
PC1 (Client)	PC2 (Server)
HI	Hello
Date	Server Date
Time	Server Time
IP	Server IP

- You can use the following method to get the local machine's IP address.
  java.net.InetAddress.getLocalHost().getHostAddress()

3. **Packet Tracer Problem Statement: Multi-Router Interconnected Network**
Objective: Design and configure a network consisting of three routers, multiple switches, and end devices (PCs and laptops) across three different subnets. Ensure full end-to-end connectivity using appropriate IP addressing, routing, and interface configurations.

Network Topology Overview:

- **Router0** connects:
  - Subnet: 192.168.1.0/24 (LAN)
  - Network: 172.16.0.0/16 (Router-to-Router communication)
- **Router1** acts as the core backbone router:
  - Interfaces to Router0: 172.16.0.1
  - Interfaces to Router2: 172.17.0.1
- **Router2** connects:
  - Subnet: 192.168.0.0/24 (LAN)
  - Network: 172.17.0.0/16 (Router-to-Router communication)



Task Requirements:

1. IP Addressing:
- Assign correct IP addresses to all router interfaces and PCs based on the diagram.

2. Routing Configuration:
- Configure all routers to ensure all networks can communicate.
- Ensure Router0 can ping PCs connected to Router2 and vice versa.

3. Device Configuration:
- Configure all PCs and laptops with appropriate **IP addresses, subnet masks,** and **default gateways.**
- Configure all routers' interfaces (using interface, ip address, no shutdown commands).

4. Connectivity Verification:
- Use the ping command to test:
  - PC0 ↔ PC1
  - Laptop0 ↔ PC2
  - Router0 ↔ Router2
- Record results and troubleshoot if any communication fails.

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